VR Graffiti



Why Learn Graffiti in VR?

Graffiti is a physical skill, not just a visual art.

Learning graffiti requires coordination between:

Movement

controlling hand trajectories and smooth strokes.

Distance & Speed

adjusting spray width, opacity, and consistency.

Lettering

understanding letter structure, proportions, and styles.

The problem

Beginners face barriers to practicing consistently, including:



Limited access to appropriate spaces



Material costs



Safety considerations

The opportunity

VR can provide an accessible, repeatable environment for learning and experimenting with graffiti techniques.

Graffiti combines artistic expression with embodied motor skills, but beginners rarely have access to structured, repeatable practice with feedback.

Existing Solutions & The Gap



Kingspray Graffiti VR

Realistic spray physics, multiple caps, multiplayer painting, and reference projection.

Tutorials focus on application features rather than graffiti techniques or artistic fundamentals.

Gap: Primarily a creative simulation tool, better suited to experienced artists than complete beginners.



Vermillion VR

Realistic oil painting, guided tutorials, reference tracing, and color mixing.

Gap: Focuses on traditional painting rather than graffiti-specific techniques and lettering.

Our Opportunity

A beginner-oriented VR learning environment with structured instruction, progressive skill development, and measurable feedback.

From Simulation to Skill Learning

Observe → Trace → Reproduce → Remix

Our learning experience draws on three educational theories:

Observational Learning

Observe and imitate an expert's hand movements through an animated virtual hand.

Scaffolding

Gradually remove visual guides and assistance as learners gain proficiency.

Constructionism

Develop understanding by creating personalized graffiti and experimenting with different styles.

From imitation to independent creation.

Learning the Tools

Different tools, different techniques.



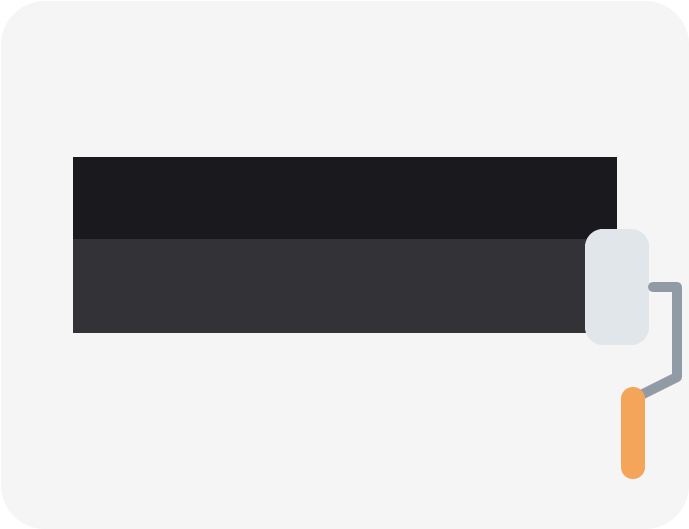
Spray Can*

Practice spray distance, movement speed, and coverage.



Marker

Practice controlled lines, sketching, and precise outlines.



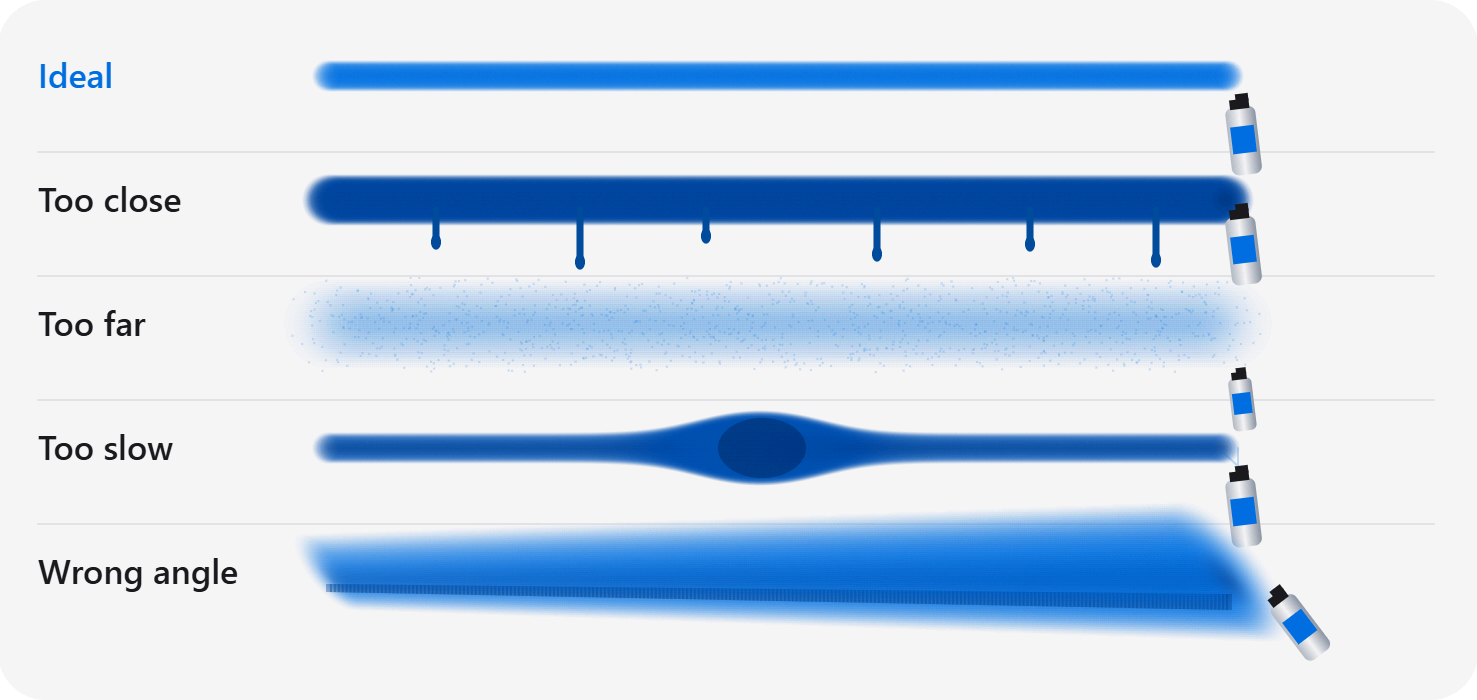
Roller

Practice broad strokes, large fills, and surface coverage.

*** Spray Can is the MVP priority.** Marker and Roller are optional, time permitting.

Practicing Stroke Techniques

Consistent lines require precise control of distance, speed, and angle.



Same stroke. Different physical control.

Precise control

Distance

Coverage & edge quality

Speed

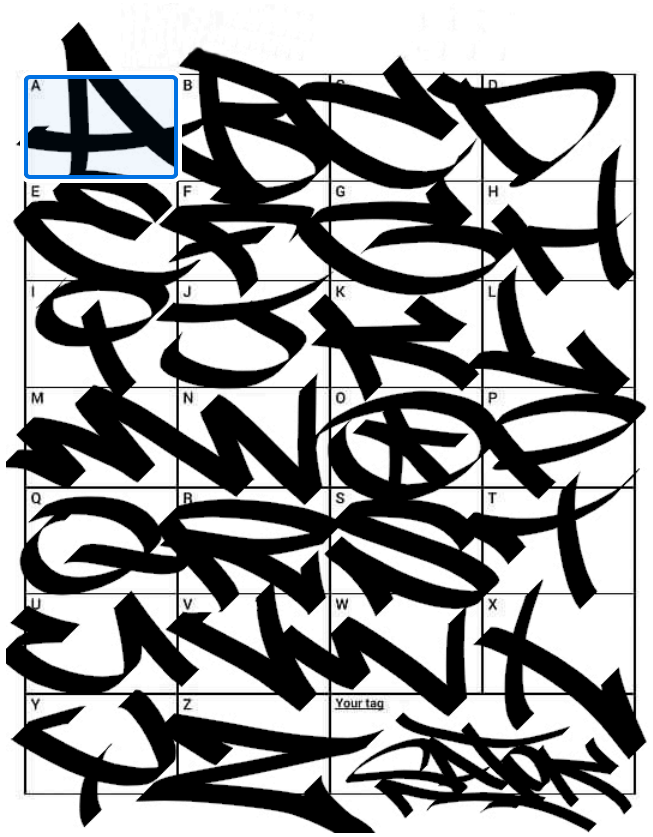
Paint buildup

Angle

Even distribution

Learning Lettering Styles

One letter at a time.

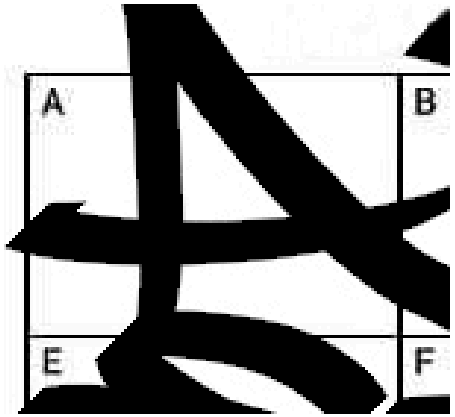


Select a letter to explore

Letter A

Bombing Science

Explore lettering styles ↗



Observe → Trace → Reproduce

Structure & proportions

Stroke order & direction

Individual letters, then words and tags

Creative Expression

A space for experimentation and personal style.



Concrete



Brick



Container

Customize the wall. Experiment with materials. Create your own style.

01 Tools → 02 Strokes → 03 Lettering → 04 Create

Technical Implementation Plan

Hardware & Software

Meta Quest 3S

Unity

C#

XR Interaction Toolkit

Core Implementation

Track movement

Track hand position, angle, speed, and distance from the wall.

Simulate spray

Simulate realistic spray behavior based on movement.

Record & replay

Record and replay expert movements for guided practice.

One More Thing...

Every Artwork Remembers

What if an artwork could remember how it was created?

VR Graffiti captures more than the final painting. It records the artist's **movements, gestures, timing, pauses, and revisions.**

These creative traces are transformed into an explorable **3D abstract artwork,** inspired by the layered, dynamic compositions of **Julie Mehretu.**

Others can enter the artwork, replay the creative process, and remix it with their own movements.



Julie Mehretu · *Stadia II*, 2004

Research Question

Can an artwork become a medium for **transmitting embodied knowledge?**